



REVIEW ARTICLE



Left Ventricular Reverse Remodeling in Heart Failure: Remission to Recovery

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ABSTRACT

Increased left ventricular dilation and reduced systolic function are consistently associated with a poor prognosis and worse clinical outcomes. In this review, we discuss the pathophysiological mechanisms, use of cardiac imaging, and clinical implications of adverse cardiac remodeling. Unique situations with evidence of spontaneous recovery and their underlying mechanisms are evaluated. A comprehensive appreciation of current evidence surrounding the array of therapeutic tools available to promote reverse remodeling and improve clinical outcomes, including medical management, revascularization, cardiac synchronization therapy, mitral valve repair or replacement, and surgical procedures, such as left ventricular assist device implantation and ventricular reconstruction, are provided. There is an ongoing need for further research identifying potential mechanistic pathways and optimal patient selection criteria for therapeutic approaches promoting reverse remodeling and myocardial recovery.

ARTICLE HISTORY Received 4 May 2021; Revised 15 June 2021; Accepted 24 June 2021

KEYWORDS Remodeling; cardiac dysfunction; heart failure

Introduction

One of the first uses of the term *remodeling* in the literature was in 1982 by Hochman and Bulkley, describing resorption of necrotic tissue, granulation tissue and scar formation after myocardial infarction (MI) in rats.¹ Although these investigators suggested it in the context of prevention of infarct expansion as a result of a healing process, the term began to be used in an adverse connotation when studies emerged demonstrating that changes in ventricular size, function, and shape were associated with decreased survival and worse outcomes.^{2,3} In the post-myocardial infarction rat model, Pfeffer and colleagues described the association of the acute and chronic changes to ventricular dilatation with consequent increased mortality. Shifts in the ventricular end-diastolic pressure–volume relationship were used to depict longitudinal alterations in post-infarction ventricular remodeling.⁴ These early findings were then translated therapeutically to show the potential of captopril in reducing ventricular dilation in rats compared to a control group with similar-sized infarcts and distending pressures.² Ten years after the introduction of the concept of ventricular remodeling in animals, Pfeffer et al. conducted the Survival and Ventricular Enlargement (SAVE) study, which demonstrated that the use of captopril in patients post MI with an EF of less than 40% was associated with improved survival as well as reduced morbidity and mortality secondary to major cardiovascular events.⁵

In 2000, an international forum published a consensus document addressing concepts and clinical implications of cardiac remodeling.⁶ The process was then defined as an assemblage of physiologic and pathophysiologic changes occurring in response to cardiac injury, pressure overload,

or volume overload through different types of insults, which share common mechanistic pathways.⁶ Adverse cardiac remodeling was classically described in the context of myocardial infarction, but changes in cardiac geometry and function can be seen with various myocardial diseases and different stimuli such as neurohormonal activation, hemodynamic changes, as well as inflammation.⁷ Cardiac remodeling can occur post myocardial infarction and myocyte loss, as well as with pressure overload associated with aortic stenosis and hypertension, inflammation associated with myocarditis, volume overload associated with valvular regurgitation, or again with genetic modifications.^{7,8} Consequently, functional, structural, biochemical, as well as cellular and molecular changes are observed as the heart remodels, a deleterious process linked with progressive ventricular dysfunction and HF decompensation.⁸ Several therapies that target the maladaptive processes central to the expression of adverse remodeling have been found to slow or reverse ventricular dilation and dysfunction, a process referred to as reverse remodeling, which is associated with improved prognosis.⁹

With this review, we will first discuss the pathophysiological mechanisms underlying cardiac remodeling and further put the emphasis on its clinical implications as well as on the imaging modalities used to predict and assess ventricular dilation and dysfunction. Subsequently, we will introduce the issue of heterogeneity in diagnostic criteria used in clinical trials for reverse remodeling, and discuss the mechanisms, biomarkers, and imaging techniques associated with regression of ventricular dilation and improvement of cardiac function. Finally, we will present existing therapies, procedures, and surgical techniques associated with proven ability to modify ventricular remodeling. Given the growing body of

evidence on the importance of reverse ventricular remodeling in HF prognosis, reversal of this complex and progressive transformation should be considered a primary treatment goal. The mechanisms of right ventricular (RV) adverse and reverse remodeling are complex and differ in many ways from remodeling processes described in the left ventricle (LV). The anatomy, embryology, and principles specific to RV dysfunction, its interaction with pulmonary pathology and their collective outcomes are outside the scope of this paper and will not be discussed here. Similarly, we shall not discuss the syndrome of heart failure with preserved ejection fraction as a specific entity in this review as we believe it to be outside the primary scope of this paper.

Cardiac remodeling

Pathophysiological mechanisms

Numerous physiological and pathophysiological changes occur in response to various mechanical and biochemical stressors, which promote the development of ventricular remodeling. These changes occur at the level of myocytes, the extracellular matrix, and the sympathetic and renin-angiotensin-aldosterone systems, with profound effects on protein function, signaling pathways, metabolic processes, and gene expression (Figure 1).

Early and late cellular changes

The progressive cellular loss secondary to any type of cardiomyocyte injury plays an important role in ventricular remodeling.⁸ Apoptosis, necrosis, and autophagy, which can together be referred to as necroptosis, play an essential role in remodeling through progressive myocyte loss.¹⁰ Recent evidence also indicates that worsening of ventricular function may be associated with changes in the process of autophagy, which can be adaptive or maladaptive.^{10–12} Specifically in the post-MI phase, early cellular changes affect both the infarcted and the non-infarcted tissue within hours to days of the initial insult.¹³ The influx of inflammatory cells and fibroblasts that ensues leads to alteration of ventricular geometry through destruction of the collagen matrix and thinning of the myocardium, as well as deposition of a new collagen matrix that contributes to scar formation in the infarcted area.^{13–15} Late cellular changes occur after weeks to months and transition from initial compensatory adaptive mechanisms to deleterious changes leading to LV dilation, increased LV end-systolic and end-diastolic volumes (LVESV and LVEDV, respectively), and ultimately LV systolic dysfunction.^{4,13,16} Hemodynamic consequences contributing to the progressive development of ischemic dilated CMP include increased load on viable myocytes, increased wall stress and preload-dependent myocardial oxygen demand, and increased LV preload in a myocardium incapable of generating sufficient contractility in response.^{13,17}

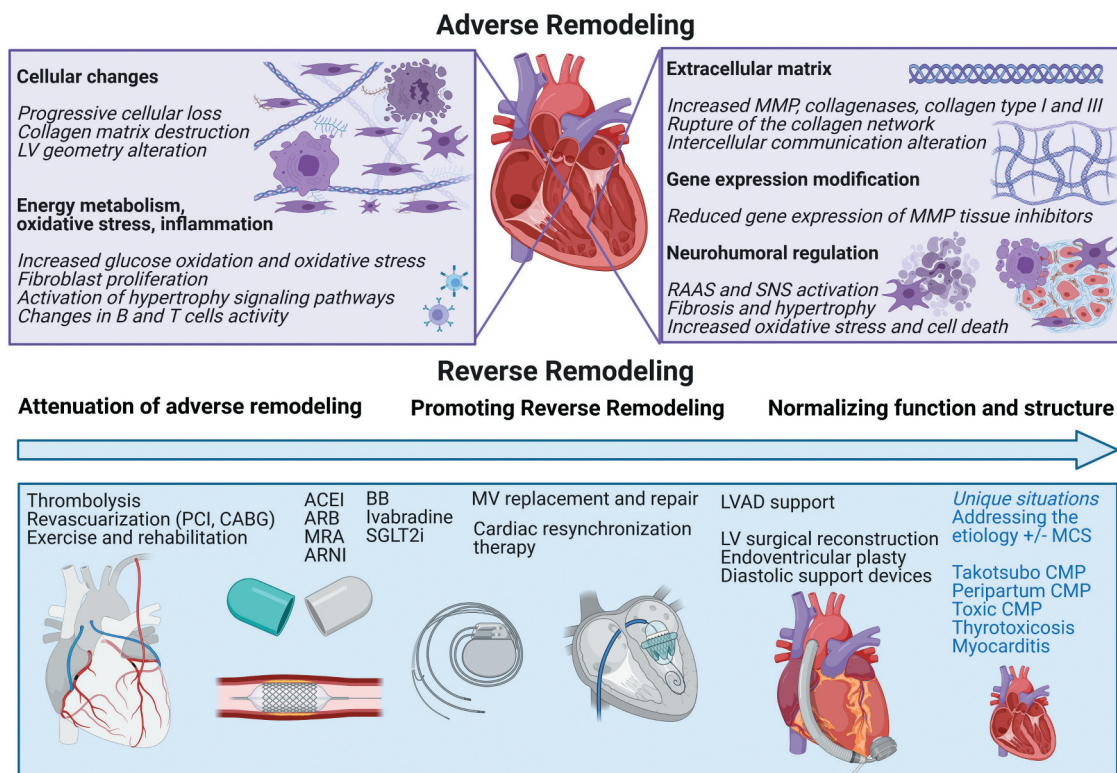


Figure 1. Pathophysiological changes associated with adverse remodeling and potential interventions to promote reverse remodeling.

ACEI: angiotensin-converting enzyme inhibitor, ARB: angiotensin II receptor blocker, ARNI: angiotensin receptor neprilysin inhibitor, BB: beta blocker, CABG: coronary artery bypass surgery, CMP: cardiomyopathy, LV: left ventricle, LVAD: left ventricular assist device, MCS: mechanical circulatory support, MRA: mineralocorticoid receptor antagonist, MV: mitral valve, PCI: percutaneous coronary intervention, SGLT2i: sodium-glucose transporter 2 inhibitor, RAAS: renin angiotensin aldosterone system, SNS: sympathetic nervous system.



Energy metabolism, oxidative stress, and inflammation

Cardiac remodeling is associated with altered energy metabolism and subsequent energy deficit with increased glucose oxidation and decreased free fatty acid oxidation. This is in contrast with normal cardiac energy metabolism in which free fatty acids account for 60% to 90% of the energy substrate for the heart.^{8,18,19} Secondary to these energy metabolism changes, accumulation of triglycerides and lipotoxicity occurs, as well as mitochondrial dysfunction and mitochondrial atrophy.⁸ In a state of imbalance between oxygen supply and consumption, these findings lead to even lower energy availability for myocardial proteins with ATPase activity, and the excessive generation of oxygen species with associated decreased antioxidant defense.^{20,21} Oxidative stress ensues and consequences are multiple, contributing significantly to cardiac remodeling through the activation of hypertrophy signaling pathways, proliferation of fibroblasts, activation of metalloproteinases, induction in apoptosis, as well as alterations in lipids, proteins, and DNA.²² Additionally, nonspecific inflammatory responses as well as B and T cells mediated specific inflammatory responses are activated secondary to cardiac injury, and on their own contribute to even more proliferation of fibroblasts, activation of metalloproteinases and loss of myocytes through apoptosis.²³ Interestingly, ventricular remodeling is also influenced through the modulation of both the innate and adaptive immune responses, particularly following ischemic cardiac injury, a principal factor associated with upregulation of inflammation in the heart.^{23–25}

Extracellular matrix and collagen

During cardiac remodeling, not only are cellular components affected, but the extracellular matrix and collagen network surrounding cardiomyocytes undergo constant turnover, affecting both the geometry and architecture of the LV.¹³ Matrix metalloproteinases (MMP) and their inhibitors, as well as collagenases, rise early after an initial ischemic insult, interfering with their usual regulation and balance within their cellular scaffold. This has been shown both in animal models and in patients post MI.^{14,26} The increased MMP activity is also responsible for progressive ventricular dilation and cardiac dysfunction post MI, with rupture of the collagen network responsible for maintaining cardiac myocardial structure alignment, regulation of muscle distensibility, and transmission of strength during protein fiber shortening.²⁷ The global geometry and shape of the LV progressively changes, moving away from a normally ellipsoidal structure to adopt a more spherical form leading to functional consequences.²⁸ Also, secondary to the remodeling process, there is increased myocardial stiffness and dysfunction secondary to the development of fibrosis with abnormal accumulation of type III and type I collagen.^{29,30}

Changes in gene expression

Compared with healthy myocardium, both non-ischemic and ischemic dilated CMP may lead to a significant increase in myocardial MMP activity and reduction in expression and activity of tissue inhibitors of metalloproteinases (TIMP), leading to myocyte reorganization and collagen matrix

breakdown.^{31,32} Modification of the extracellular matrix (ECM) secondary to altered MMP and TIMP gene expression may also influence intercellular communication through the ability of the ECM to influence extracellular cytokines and signaling proteins via the matricellular proteins. These include osteopontin, osteonectin, thrombospondins, tenascin, and periostin, which will interact with cell-surface receptors and growth factors, linking ECM proteins together and modulate cell behavior.³³ The activity of matricellular proteins will be enhanced in response to cardiac injury or other neurohormonal stressors.³³

Neurohumoral regulation

The renin-angiotensin-aldosterone and sympathetic nervous systems (RAAS and SNS) are critical in regulatory pathways of cardiac remodeling and activation of both systems stimulate protein synthesis in fibroblast and cardiomyocytes, leading to fibrosis and hypertrophy.³⁴ They also contribute to activation of MMP and increases in oxidative stress and cell death through apoptosis and necrosis, in addition to their effect on salt and water retention and vasoconstriction, which will be discussed further in the following section.^{35–37} Angiotensin II independently promotes cardiac remodeling through increased perivascular and interstitial collagen content as well as direct cytotoxic effects on cardiomyocytes promoting apoptosis and cellular hypertrophy.³⁸ The changes modulated through angiotensin II can be directly enhanced by chronic SNS activation which can, on its own, promote cardiomyocyte apoptosis.³⁹ The chronic catecholamine activity that ensues can directly promote fibrosis, oxidative damage, and cardiac dysfunction.⁴⁰

Mechanical implications of cardiac remodeling

Cardiac remodeling may be divided into adaptive and maladaptive remodeling.^{6,41} Adaptive remodeling can represent physiologic changes in proportions and function of the heart in response to stimuli such as pregnancy or exercise, but it can also be the initial apparent compensatory process to various pathological triggers, which enables the heart to maintain its normal function.^{6,42} Indeed, cardiac remodeling may occur in response to pressure overload secondary to hypertension or aortic stenosis, volume overload secondary to valvular regurgitation, cardiac injury secondary to MI, myocarditis, or again idiopathic dilated CMP.⁴² The first hemodynamic changes of adaptive remodeling are regulated by the SNS and the RAAS. The increase in β -adrenergic tone and activation of the RAAS responsible for regulation of extracellular volume, sodium balance, as well as cardiac and vascular trophic effects, aim at increasing and maintaining cardiac output by their effects on heart rate and stroke volume.^{34,43} By comparison, persistent SNS and RAAS activation leads to maladaptive and deleterious remodeling, giving rise to progressive cardiac dysfunction, which is associated with poorer outcomes.^{8,44,45} Different phenotypes in cardiac remodeling result from distinctive diseases that share common pathways in terms of molecular, biochemical, and mechanical events.⁶ Secondary to myocardial infarction, reparation of the necrotic area and scar formation may initially be considered beneficial, but



progressive LV dilatation, enhanced hemodynamic burden, and ventricular geometry distortion lead to worsening myocardial function over time.^{6,46} With pressure overload secondary to hypertension or aortic stenosis, cardiomyocytes enlarge progressively leading to ventricular concentric hypertrophy and stiffening.¹⁰ The initial adaptive response of increased ventricular wall thickness and ventricular dilation diminishes wall stress and decreases oxygen demand, but with persistent pressure overload, a transition to maladaptive mechanisms with decompensation and clinical HF occurs. Even within the same disease such as severe aortic stenosis, several patterns of remodeling patterns are observed, and it is increasingly recognized that the process is heterogeneous and less well predictable.⁴⁷ With volume overload such as with primary MR, the LV adapts and changes its global structure with eccentric hypertrophy to compensate and produce an adequate stroke volume. The hypertrophy eventually becomes insufficient and progressive myocardial dysfunction develops.^{48,49}

Electrical implications of cardiac remodeling

The association between malignant ventricular arrhythmias and cardiac remodeling has been established through various well-defined mechanisms. Remodeling is associated with prolongation of the action potential, changes in sodium, potassium, and calcium channels, including alterations in the sodium/calcium exchanger.^{50,51} It also disturbs the gap junctional intercellular communication as evidenced by modification of the normal distribution and labeling intensity of connexin 43, the most prominent gap junction protein expressed in the normal heart.^{50,51} These alterations may be associated with prolongation of the QT interval and promote early after depolarizations and increase the risk of malignant arrhythmias.⁵⁰ Remodeling also causes the accumulation of myocardial fibrillar collagen involving the epimysium, perimysium, and endomysium, three connective tissue compartments surrounding muscle bundles (epicardium and endocardium), muscle fibers, and single muscles, respectively. The development of fibrosis through increased collagen content may lead to electrical conduction disorders and arrhythmogenesis, especially reentrant arrhythmias facilitated by activation delays with electrogram fractionation through fibrotic muscles.^{50,52} Altogether, those changes associated with cardiac remodeling increase the risk of sustained ventricular tachycardia and ventricular fibrillation, thus raising patients' risk of sudden cardiac death.^{8,53}

Cardiac imaging

Cardiac imaging is currently the gold standard for proper assessment of resting LV function and infarct size after myocardial infarction, where transthoracic echocardiogram (TTE) is recommended within the first 24 to 48 hours after a MI.^{54,55} TTE can also help with the identification of patients at increased risk of mortality and poorer outcomes through volume measurements. A LVEDV increase of more than 20% has been found to be associated with increased mortality in patients post MI after receiving percutaneous coronary

intervention.⁵⁶ Additionally, in echocardiographic sub-studies of the Valsartan in Acute Myocardial Infarction Trial (VALIANT) trial, abnormal LV geometry increased the risk of major cardiac events, larger infarct size, and systolic dysfunction were associated with higher all-cause mortality, and LVESV as well as LVEDV were associated with adverse clinical events.^{57,58} In the same way, a greater change from baseline post MI to one-year LV systolic area and percentage change in cavity area were strong predictors of cardiovascular mortality and adverse cardiovascular events in sub-studies of the SAVE trial.⁵⁹ Similar observations were obtained from the Val-HeFT trial linking worse LV internal diastolic diameter and lower EF to higher risk of cardiac events.⁶⁰ Other echocardiographic tools such as global longitudinal strain (GLS) are valuable for LV remodeling assessment as it is more sensitive than LVEF to assess myocardial dysfunction and is more reproducible than LVEF.⁶¹ Some studies have also used contrast ventriculography to assess cardiac remodeling after MI and found LVESV to be a strong predictor of 6-month mortality and to portend a greater predictive value for survival than LVEDV and LV systolic function.^{44,62}

Cardiac magnetic resonance (CMR) imaging allows for more accurate tissue characterization and has begun to gain interest as a tool to assess ventricular remodeling in a variety of cardiovascular conditions to describe with more precision the extent and location of fibrosis.^{55,63,64} In addition, the use of late-gadolinium enhancement (LGE) adds the possibility of identifying the timing of infarction as well as the determination of myocardial injury reversibility.^{65,66} Studies conducted in ischemic and nonischemic cardiomyopathy (CMP) demonstrated the prognostic importance of T1 mapping used to assess myocardial fibrosis.^{67–69} In nonischemic CMP, Puntmann et al. demonstrated that T1 mapping was a strong predictor of all-cause mortality and HF events.⁶⁷ While LVEF and LV volumes have traditionally been used to evaluate cardiac remodeling, other imaging tools may allow for a more detailed assessment and provide better tissue characterization by evaluating the structural aberrations that underlie ventricular remodeling.

Reverse remodeling

Cardiac remodeling was initially believed to be an irreversible process with permanent architectural and functional changes. Many studies in the past few decades have shed light on the capability of the heart to reverse the consequences of maladaptive mechanisms, and that cardiac structure and function can improve to a certain extent with the help of optimal medical therapy and devices. Reverse remodeling is a dynamic process that can be seen in diverse clinical settings. It is associated with better clinical outcomes and improved prognosis; a continuum during which different mechanisms are set in motion to attempt reverting the heart closer to normal geometry and function.²⁸ The process of reverse remodeling intertwined with the concept of myocardial recovery has yet to be fully understood and a standardized definition is lacking. The heterogeneity in cutoff thresholds used in research studies is illustrated in the current literature, with the evaluation of mixed patient populations and use of different



imaging techniques. Various reverse remodeling criteria have been used, both with TTE and CMR. Examples of such criteria include a relative percent change of LVESV index of 15% or more, absolute LVEF increase of at least 10% or 15%, relative percent change of LVEDV index of 10% or more, specific cutoff of LVEDV index of 33 mm/m² to be reached, relative LVEDV reduction or at least 10%, relative LVESV index reduction of 20% or more, or again relative LVESV reduction of at least 40%.^{70–75} Some studies have also included absolute EF cutoffs to be reached as part of their composite criteria,^{70,76} or again LV mass reduction as a continuous variable to assess for reverse remodeling.⁷⁷ Those examples illustrate the lack of a standardized definition to properly assess reverse remodeling and more specifically to better define the response to modern guideline-directed therapy.

Therapeutic interventions that have been shown to intervene with the process of adverse remodeling and/or to promote reverse remodeling are shown in **Figure 1** and include but are not limited to coronary revascularization, pharmacological interventions, and device-based or surgical options, such as CRT, LVAD, and valvular procedures. **Figure 2**^{78–95} and **Figure 3**^{87,96–108} depicts the generally expected improvements in LVEF and LV size with these therapeutic interventions based on clinical trial evidence (note should be made that these changes reflect sequential benefits of each intervention on top of prior therapy, in most cases). While they are numerous and diverse, the consequences of these therapies can roughly be directionally classified into three trajectories of outcome, which will be discussed here. Some therapies may attenuate the progression of adverse remodeling. Others may improve but not restore structure and function by their effects at the clinical, structural/functional, and/or cellular level. Finally, some therapies may result in restoration of the clinical and structural/functional HF phenotype, such that recovery is achieved.

Attenuating the progression of adverse remodeling

Therapeutic interventions that alter the progression of adverse remodeling have been principally developed to target neurohormonal activation and to promote reduction of infarct extent after MI. As such, in patients post acute MI, the use of early thrombolysis has been shown to be beneficial in limiting the extent of regional wall motion abnormalities and to reduce the degree of ventricular dilation with smaller LVESV and LVEDV on TTE at 6-months post MI.^{109,110} Apart from infarct size, late patency of the culprit artery was found to be a major determinant of left ventricular volume change and function after infarction.^{111,112} As discussed previously, the demonstration that ACEI use was slowing adverse cardiac remodeling through LV dilation attenuation and improved survival after MI was first established in the rat model,¹ and subsequently in patients following Q-wave MI with LV dysfunction.¹¹³ Thereafter, a number of medical therapies demonstrated clinical and functional benefits within the arrest of adverse cardiac remodeling to promotion of reverse remodeling pathophysiological continuum, which will be discussed in more depth in the following section.

Exercise training and cardiac rehabilitation have also been proposed as other strategies to slow or arrest adverse cardiac remodeling. Mechanistically, exercise training achieves a sympatho-inhibitory effect, which serves to de-escalate neurohormonal activation in heart failure.^{114,115} Clinical data however, while reassuring of the general safety of exercise and rehabilitation programs in HF patients, offers mixed evidence as to their efficacy with regard to adverse remodeling. In a 2019 Cochrane review on exercise-based cardiac rehabilitation in HF, while the intervention did not reduce the risk of all-cause mortality, there were improvements in all-cause and HF-related hospitalizations as well as functional scores.¹¹⁶ Improvements in LV parameters remain controversial with at least one older meta-analysis favoring aerobic over strength training in improving LVEDV and LVESV, though with varying exercise regimens.¹¹⁷ A study using CMR in patients with HFrEF demonstrated neutral effects on LV parameters of a 2-month aerobic interval regimen, highlighting the need for longer term data with standardized exercise programs.¹¹⁸

Promoting reverse remodeling

Revascularization

In patients post MI, early revascularization has been associated with better outcomes, smaller infarct size, and improved LV function. Studies have shown that primary percutaneous coronary intervention after ST-segment-elevation MI was associated with reverse remodeling, defined as a reduction of more than 10% in LVESV at 6 months, was in turn associated with a decreased incidence in HF, favorable long-term outcomes, as well as improved survival.^{119,120} Coronary artery bypass surgery has also demonstrated the ability to promote ventricular remodeling with LVEF, LVEDV, and LVESV improvement within 3–6 months post-operatively.^{121,122} In a study conducted in patients with decreased LV function undergoing surgical revascularization with bypass surgery, lower LVESV index was the only independent predictor of improved survival.¹²³ Compared to patients receiving medical therapy alone, patients undergoing coronary artery bypass surgery have significantly lower rates of death from any cause, death from cardiovascular causes, and death from any cause or hospitalization for cardiovascular causes over 10 years.¹²⁴

Medical therapies

A number of medical therapies have been shown to promote reverse remodeling with restoration of a more normal ventricular shape, reduction in LV volumes and mass, as well as an improvement in LVEF. These architectural and functional changes are linked with reductions in morbidity and mortality.¹²⁵ In the post-MI population, the early use of ACEI was found to be associated with improved short-term and long-term morbidity and mortality, with improvements in all-cause death apparent as early as 30 days after the event, notably in the Survival and Ventricular Enlargement (SAVE) trial echoing the results of the CONSENSUS and Studies of Left Ventricular Dysfunction (SOLVD) trials.^{5,126–131} In the one hundred and eight HFrEF patients enrolled in the

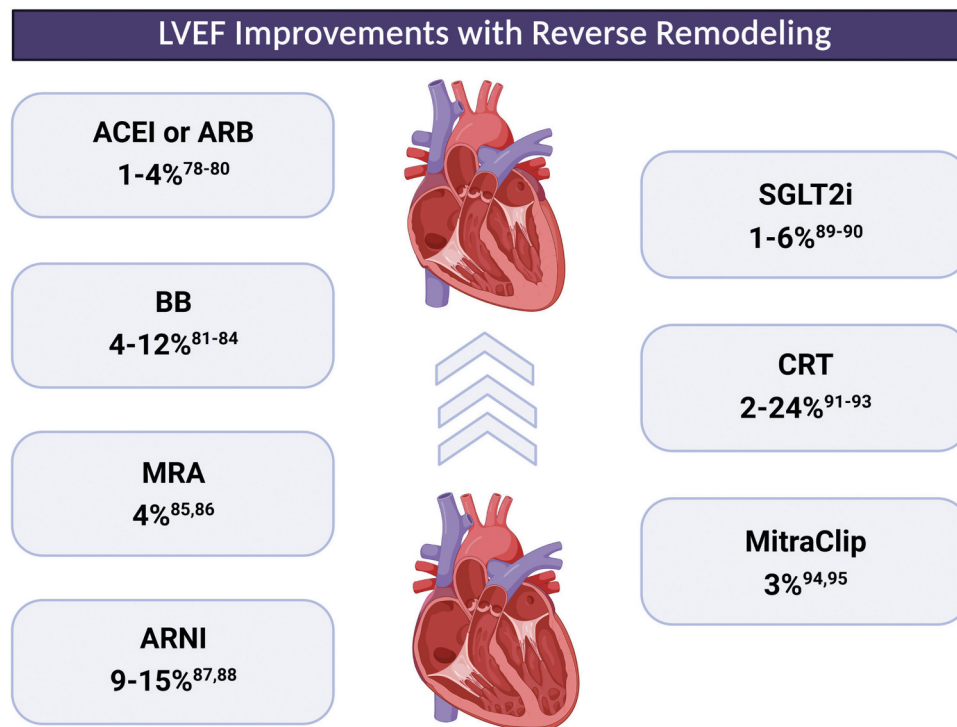


Figure 2. Left ventricular ejection fraction improvements expected with therapeutic interventions targeting reverse remodeling. (The left ventricular ejection fraction improvements seen with angiotensin-converting enzyme inhibitor, angiotensin II receptor blocker, and beta blockers represent “stacked on” benefits).

ACEI: angiotensin-converting enzyme inhibitor, ARB: angiotensin II receptor blocker, ARNI: angiotensin receptor neprilysin inhibitor, BB: beta blocker, CRT: cardiac resynchronization therapy, LVEF: left ventricular ejection fraction, MRA: mineralocorticoid receptor antagonist, SGLT2i: sodium-glucose transporter 2 inhibitor.

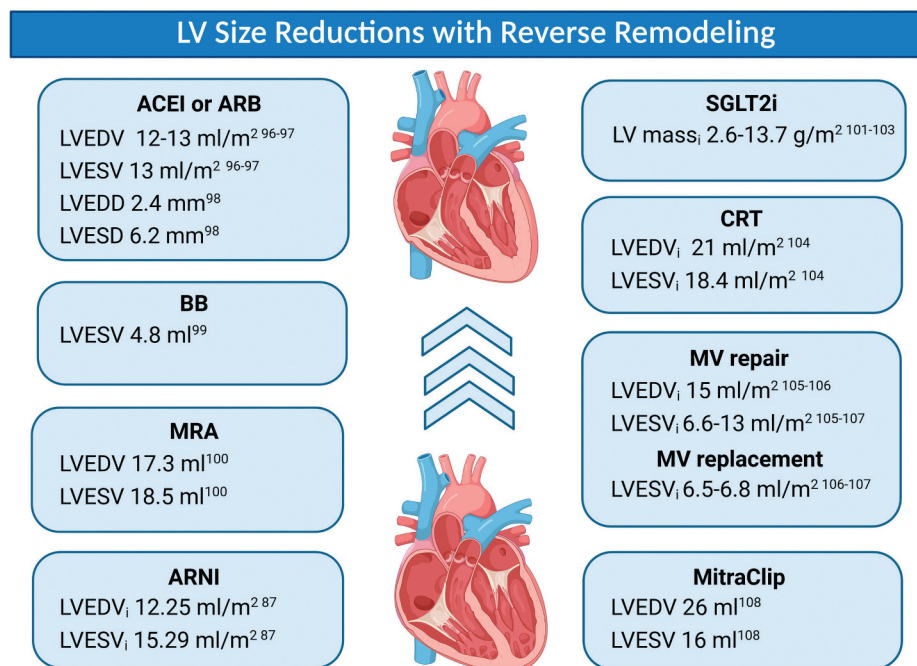


Figure 3. Left ventricular size reductions expected with therapeutic interventions targeting reverse remodeling. (The left ventricular ejection fraction improvements seen with angiotensin-converting enzyme inhibitor, angiotensin II receptor blocker, and beta blockers represent “stacked on” benefits).

ACEI: angiotensin-converting enzyme inhibitor, ARB: angiotensin II receptor blocker, ARNI: angiotensin receptor neprilysin inhibitor, BB: beta blocker, CRT: cardiac resynchronization therapy, LV: left ventricle, LVEDD: left ventricular end-diastolic diameter, LVEDV: left ventricular end-diastolic volume, LVEDV_i: left ventricular end-diastolic index, LVESD: left ventricular end-systolic diameter, LVESV: left ventricular end-systolic volume, LVESV_i: left ventricular end-systolic volume index, MRA: mineralocorticoid receptor antagonist, MV: mitral valve, SGLT2i: sodium-glucose transporter 2 inhibitor.



SOLVD trial, radionuclide ventriculogram before and after double-blind randomization to enalapril or placebo demonstrated a significant reduction in LVEDV at a mean of 25 months in asymptomatic patients.⁹⁶ In symptomatic HFrEF patients, there was a significant reduction in LVESV and LVEDV, with concomitant increase in mean LVEF at 1 year.⁹⁷ Statistically equivalent improvements in clinical outcomes were observed with the use of angiotensin receptor blocker (ARB) compared to ACEI in post-MI patients over a median of 24.7 months in the VALIANT trial, and after 6 months in the OPTIMMAL trial.^{132,133} After acute MI, there was no difference between valsartan and captopril at 20 months with respect to improvement in LVEF and reduction in LVESV and LVEDV.⁵⁷

Mineralocorticoid receptor antagonists (MRA) have been shown to improve outcomes across the spectrum of HFrEF, including patients post MI with clinical evidence of HF. In the Randomized Aldactone Evaluation Study (RALES), spironolactone was associated with significant lower risks of death from any cause, sudden cardiac death, and death due to progressive HF at 24 months. Also, Eplerenone Post Acute Myocardial Infarction Heart Failure Efficacy and Survival Study (EPHESUS) demonstrated that eplerenone use in patients post MI and LV dysfunction was associated with reduced time to death from any cause and the composite time to cardiovascular death or first cardiovascular hospitalization after a mean follow-up of 16 months.¹³⁴ In patients with LVEF $\leq 35\%$ with New York Heart Association class II symptoms, eplerenone significantly reduced cardiovascular death or heart failure hospitalization, as well as all-cause mortality at 21 months of median follow-up in the EMPHASIS-HF trial.¹³⁵ MRA have been shown to improve LVEF, LVESV, and LVEDV after 6 and 12 months of therapy.¹⁰⁰ Angiotensin receptor-neprilysin inhibition (ARNI) therapy has been compared to standard HF therapy with ACEI and has been found to decrease HF hospitalizations and improve survival after a median follow-up of 27 months in the PARADIGM-HF trial.¹³⁶ Subsequently, the PROVE-HF prospective observational study sheds light on the ability of ARNI to promote ventricular remodeling at 12 months in patients with HF.⁸⁷ Among close to 800 HFrEF patients, there was a statistically significant 9.4% increase in mean LVEF along with clinically significant reductions in LVEDV and LVESV indexes, 12.25 ml/m² and 15.29 ml/m², respectively. These changes are particularly impressive since most enrolled patients were already exposed to ACE inhibitors or ARBs.

Improved survival and clinical outcome, as well as positive architectural LV changes in the HF population, has consistently been shown with the use of BB therapy.^{137,138} It was especially true in patients post MI with significant relative risk reduction of all-cause mortality with the use of carvedilol compared with placebo.¹³⁹ Evidence of reduced cardiac remodeling, including improved systolic function and reduced LVESV, has been seen with carvedilol compared to controls at 6 months post MI, even in the presence of significant ACEI use in both groups.⁹⁹ Additionally, the heart rate lowering medication ivabradine was shown to induce reverse remodeling in patients with HF in

a substudy of the (SHIFT) trial demonstrating an absolute reduction in HF mortality or hospitalization at 2 years in patients with a LVEF $\leq 35\%$ on contemporary medical therapy with a resting heart rate ≥ 70 beats per minute.^{140,141} After 8 months of treatment, LVESV and LVEDV indexes, as well as LVESV and LVEDV were significantly reduced in patients taking ivabradine compared to placebo. An increase in LVEF was observed concomitantly over the same period of time. Evidence of reverse remodeling with ivabradine was also observed in a case-controlled study evaluating its effect on LV dyssynchrony parameters in HF patients.¹⁴² LVESV reductions were observed as early as 1 month after initiation of therapy compared to placebo, but more pronounced at 3 months, as well as statistically significant improvement in echocardiographic LV dyssynchrony parameters.

Evidence is progressively emerging regarding the potential mechanisms of action as well as the impact of sodium-glucose cotransporter 2 inhibitors (SGLT2i) on reverse ventricular remodeling. A few studies have evaluated the effect of the different SGLT2i on LV structure and function with both TTE and CMR. TTE performed after 3 and 6 months of SGLT2i therapy demonstrated improved LV diastolic function (through lateral e', E/e' ratios, and E/A ratio) and reduced LV mass index in patients with type 2 diabetes with known cardiovascular disease or HF with preserved and reduced EF.^{101–103,143,144} One study using dapagliflozin in 53 patients with type 2 diabetes and stable HFpEF or HFrEF found a minimal but statistically significant improvement in LVEF after 6 months of therapy.¹⁴³ Only one study evaluating the effect of empagliflozin compared to placebo in type 2 diabetes patients at 6 months was able to demonstrate improved LVEDV using CMR.¹⁴⁵ The recent EMPATROPISM trial demonstrated that empagliflozin use in nondiabetic HFrEF patients significantly improved LVESV and LVEF at 6 months.⁸⁹ The evidence is growing regarding the effect of SGLT2i on reverse remodeling and preservation of cardiac function in patients with HF with or without concomitant diabetes, and future larger trials are ongoing that will provide helpful insights into the benefits of SGLT2i in the HF population.

Cardiac resynchronization therapy

Survival, HF events, symptoms, and quality of life have all been shown to be improved in patients undergoing CRT,^{93,146–148} although the challenge remains regarding optimal patient selection. There is evidence that CRT improves ventricular function and structure in American College of Cardiology/American Heart Association stage C HF patients with greater improvement in LVESV and LVEDV indexes as well as LVEF compared to controls with no resynchronization therapy after a 12-month follow-up.¹⁰⁴ No single echocardiographic parameter has been identified to better guide the selection of better CRT responders, and up to 35% of HF patients may not benefit from CRT implantation as per the current guidelines.⁷³ Other imaging techniques and measurements such as single-photon emission computed tomography (SPECT) and viability testing could improve patient selection as it was shown that the extent of scar tissue and viable myocardium in patients with ischemic



CMP seems to be directly proportional to CRT response.¹⁴⁹ Using SPECT could identify patients in whom CRT might induce and facilitate reverse remodeling within viable muscle.

Mitral valve repair and replacement

Secondary (functional) mitral valve regurgitation is often a consequence of cardiac remodeling with dilation of the failing ventricle and progressive displacement of the papillary muscle with eventual mitral valve coaptation defect.⁴⁸ The added volume load that ensues contributes to further cardiac remodeling, thus creating a vicious cycle leading to worsened cardiac architecture and function.⁴⁸ After transcatheter edge-to-edge mitral valve repair for severe functional mitral regurgitation, significant improvement in LVEF and ventricular shape has been observed, in line with reductions in LVEDV seen at 30 days and which persist at 6-month follow-up.¹⁰⁵ Surgical mitral valve repair has been compared to surgical replacement in patients undergoing CABG and similar reductions in LVESV at 1 and 2-year follow-up were observed, with no significant change in both groups, and similar mortality rates.^{106,107} The more recent use of transcatheter edge-to-edge repair with the MitraClip for moderate-to-severe functional mitral regurgitation in HF patients who remain symptomatic despite guideline-directed medical therapy has been shown to lower mortality, HF hospitalizations, and improve functional capacity as well as quality of life within 2 years of follow-up compared to medical therapy alone.¹⁵⁰ Evidence of reverse remodeling with MitraClip was demonstrated in a retrospective study of high surgical risk patients with severe functional mitral regurgitation who had significant LVESV and LVEDV reductions at 1 year compared with controls on medical therapy alone.¹⁰⁸

LVAD support

Levin et al. established that *reverse remodeling* was possible with the demonstration that end-stage failing hearts could revert back to a normal end-diastolic pressure–volume relationship after a period of hemodynamic unloading with MCS using left ventricular assist devices (LVADs).¹⁵¹ Among all therapeutic options for HF patients, LVAD support is associated with the largest degree of reverse ventricular remodeling through not only mechanical unloading, but also with improvement in cardiac output and blood pressure, as well as improvement in neurohormonal aberrations.¹⁵² The use of LVAD support in patients with advanced HF leads to a clinically meaningful survival benefit and significantly improve their quality of life.¹⁵³ Even with the most advanced stages of HF, LVAD support has been associated with positive LV shape changes, with LV dilation reversal and normalization of ventricular geometry.¹⁵¹ The degree of reverse remodeling can be significant enough in some patients that consideration of LVAD explantation becomes possible.¹⁵⁴ Positive changes may be seen within the first month of mechanical unloading with significant improvements in LVEF, LVED diameter, and LV mass compared with pre-implantation values.¹⁵⁵

Despite clinical cardiac recovery and improvement in LV function and structure seen with reverse remodeling,

abnormal gene expression changes associated with adverse remodeling may persist, and new gene expression modifications may develop secondary to reverse remodeling. For example, the expression of more than 3000 genes has been found to be modified in response to HF in a transcriptome analysis of 199 myocardial biopsies taken from patients with failing and non-failing hearts, as well as patients with LVAD support.¹⁵⁶ Only around 5% of those genes showed normalization in their expression levels after mechanical unloading, and a new subset of genes was found to be dysregulated in this patient subgroup compared with controls. Additionally, regulators of gene expression include protein-coding and non-coding RNAs and are major contributors to molecular changes seen with remodeling.¹⁵² They are crucial to the heart's response to hemodynamic stressors and insults, and can be upregulated secondary to HF, but conflicting results exist with regard to their response to mechanical unloading.^{157–159} Different patterns of microRNA expression levels have been described in the failing heart in response to LVAD implantation. Some microRNA expression levels have been found to be decreased at the time of implantation in patients with LV recovery compared to patients with no evidence of recovery, while some specific microRNAs remained unchanged after LVAD support.^{158–160} Also, long non-coding RNAs have been found to have the most dynamic expression changes in response to mechanical unloading with MCS, but the functional significance of those changes remains unknown.¹⁵⁹ Interestingly, normal myocardial samples and samples taken from LV-recovered myocardium share similarities regarding specific microRNA levels.¹⁶⁰ Also, other cardiac-specific microRNA levels were found to decrease significantly after only a few months post LVAD implantation, potentially reflecting the heart's response to mechanical unloading.¹⁶¹ The discoveries of transcriptional abnormalities observed through fluctuating patterns of microRNAs and long non-coding RNAs may open new avenues for gene discovery and potential novel therapeutic strategies. MicroRNAs and long non-coding RNAs may be helpful tools to assess the severity of heart failure, but their contribution and significance as potential markers of myocardial recoverability has yet to be elucidated.

Importantly, critical adaptations to hemodynamic and neurohormonal signals occurring with cardiac remodeling within the ECM's composition and structure do not necessarily revert to a normal state and architecture with reverse remodeling. As the heart remodels, the turnover and stability of the ECM is influenced by numerous factors, including disease etiology, enzymatic changes, as well as neurohormonal and hemodynamic changes leading to fluctuations in ventricular wall stress.¹⁵² Notable attention has been given to collagen content with the study of myocardial remodeling with respect to the ECM's structural function. A few studies have shown that LVAD implantation is associated with decreased total collagen deposition on paired myocardial samples from patients with CMP of different etiologies,^{162,163} although there were conflicting results between patients under LVAD support whether they were on ACEI or not in the study by Klotz et al.¹⁶² Total collagen content increased in patients not taking an ACEI, while it decreased in patients taking an ACEI.



One study also showed that ECM volume was initially increased in response to LVAD implantation, with subsequent volume reduction with longer support.¹⁶⁴ Other components of the ECM have been less extensively studied, such as MMPs and TIMPs involved in the adverse remodeling process as described previously. Mechanical unloading with LVAD support does not seem to reverse levels of MMPs and TIMPs, but changes toward nonfailing levels have been described in patients with both LVAD and ACEI therapy.¹⁶² Matricellular proteins in the ECM involved in communication and signaling between cells are influenced by MCS, with restoration of the fibrillar ECM, but non-fibrillar protein abnormalities disturbing the processing and assembly of mature collagen fibrils persist after mechanical unloading.¹⁶⁵ Thus, the reverse remodeled heart continues to have abnormal features, and the functional roles of ECM biochemical and structural changes remain poorly understood in reverse remodeling.

Other surgical therapies

Surgical therapy for substantial ventricular scar tissue or significant dyskinetic and akinetic areas after MI can be considered via techniques of LV chamber surgical restoration. Endoventricular circular patch plasty technique (Dor procedure) can restore a closer to normal LV geometry and, when successful, leads to improve LV regional and global systolic function.¹⁶⁶ Surgical ventricular restoration can promote reverse remodeling through the reduction of wall stress in other myocardial areas, improving both the morphology and hemodynamic parameters of the LV.¹⁶⁷ Over 1000 patients who underwent LV surgical restoration were included in a large multicenter study, which demonstrated an overall mortality of 5.3%, a significant decrease in mean LVESV index from 80 to 56 ml/m², and a close to 10% improvement in mean EF 30 days post-operatively.¹⁶⁸ At 5 years, mean survival was 68% with a significant improvement in functional capacity. The Surgical Treatment for Ischemic Heart Failure (STICH) trial randomized patients with ischemic CMP to CABG surgery alone or CABG associated with surgical ventricular restoration.¹⁶⁹ There was a significant reduction in LV volume with a reduction in LVESV index by 19% after CABG associated with surgical restoration, compared with a 6% reduction after CABG alone at a median follow-up of 48 months. Despite this difference in LV dilation reversal, improvements in symptoms, exercise tolerance, and rates of death or hospitalizations for cardiac causes were found to be similar in the two groups. Other available surgical procedures include diastolic support devices such as the Acorn CorCap Cardiac Support Device, a biocompatible support positioned around the ventricles to prevent further dilatation.^{170,171} In a series of 48 patients, 33 patients received the Acorn CorCap Cardiac Support Device along with concomitant cardiac surgery and 15 patients received the device alone.¹⁷¹ There was a statistically significant reduction in LVEDD (71.9 ± 4.9 mm vs 63.7 ± 4.2 mm, $p < 0.05$ for device only and 72.1 ± 8.9 mm vs 62.6 ± 14.3 mm, $p < 0.05$ for device and surgery) and increase in LVEF ($22.0 \pm 4.6\%$ vs $29.8 \pm 7.4\%$, $p < 0.05$ for device only and $24.5 \pm 11.0\%$ vs $33.2 \pm 15.3\%$ for device and surgery) at 12-month follow-up. Evidence of reverse remodeling persisted at 18 and 24-month follow-up. In general, surgical reconstruction of the LV or use of passive cardiac

restraint with devices such as the Acorn CorCap device have fallen out of clinical favor.

Normalizing ventricular function and structure

Unique situations

Despite the capability of medical and device therapy, including LVAD support, to promote reverse remodeling, the rate of recovery toward a clinically and structurally normal cardiac phenotype remains low. A few unique and specific situations show evidence of spontaneous recovery, which are associated with complete reverse remodeling once the inciting adverse event or agent is resolved or removed, even in the presence of severe cardiac dysfunction and fulminant cardiogenic shock. Such situations include takotsubo CMP, peripartum (PP) CMP, CMP related to some cardiotoxic exposure (chemotherapy, alcohol), thyrotoxicosis, and myocarditis.^{172–176} Reverse remodeling and LV function recovery in PP CMP varies geographically, with previous studies reporting recovery rates as low as 21% to 29% in Africa and Haiti between 6 months and 2 years after follow-up,^{177–180} but as high as 71% in North America at 1-year follow-up¹⁸¹ and 96% in Germany at 5-years follow-up.¹⁸² Persistent LV dysfunction at follow-up has been associated with a few clinical predictors, including diagnosis during pregnancy, LVEF of 30% or less, LV end-diastolic dimension of 6 cm or more with fractional shortening of less than 20%, concomitant right ventricular dysfunction, and elevated troponin levels.^{183–187} Other subtypes of non-ischemic cardiomyopathy showing spontaneous reverse remodeling caused by “abnormal energetics” include stress-induced CMP and tachycardia-induced CMP.¹⁸⁸ In alcohol-induced CMP, most patients attain significant recovery with achievement of total alcohol abstinence, but this is rarely the case for patients with very severe LV dysfunction where the direct toxic effect of alcohol induces irreversible cardiomyocyte loss and replacement by subendocardial and interstitial fibrosis.^{189–192} The development of CMP secondary to hyperthyroidism results from an untreated high output state that leads to ventricular dilation, tachycardia, and eventual HF symptoms.¹⁹³ In Grave’s thyrotoxicosis, there may be direct damage and disturbance of apoptosis through an auto-immune form of myocarditis in addition to an underlying hyperdynamic state.¹⁹⁴ Also, hypothyroidism may lead to the development of HF with LV systolic and diastolic dysfunction, with associated bradycardia, impaired systemic vascular resistance, and endothelial dysfunction.¹⁹³ Reversal of clinical symptoms as well as improvement of echocardiographic measurements showing regression of myocardial mass can be achieved with appropriate treatment and restoration of euthyroidism, especially in patients with short disease duration and concomitant beta-blocker (BB) therapy.¹⁹⁵ Myocarditis is an inflammatory disease of the myocardium most commonly secondary to a viral etiology, but can but secondary to any form of injury to the heart, with variable clinical severity, from subclinical to fulminant presentations.¹⁸⁸ Fulminant myocarditis has a reversible nature with improved long-term survival compared with



more indolent forms of myocarditis, when managed with appropriate aggressive hemodynamic support.^{173,175} To this day, it remains challenging to define the natural history of reverse remodeling in myocarditis, since histological evidence is often lacking or equivocal, and myocarditis definitions vary between studies. Nevertheless, patients with viral myocarditis have a variable prognosis, with a reported 40% to 88% complete LV function recovery rate described across studies.^{173,196,197} When “caught in time” and managed appropriately, those unique situations with abnormal bioenergetic pathways, inflammation, or again toxic insults demonstrate a remarkable ability to recover, but their natural history is not fully understood nor is it well characterized. The nature of the inciting injury and the extent of underlying myocardial damage seem like crucial determinants of the ability of the heart to remodel and/or recover.

Remission to recovery

Although significant reverse remodeling is observed with medical therapy, devices, and surgical options including LVAD implantation, only rarely is complete myocardial recovery attained. The 2020 Society of Thoracic Surgeon-Interagency Registry for Mechanically Assisted Circulatory Support described a 4-year rate lower than 5% of LVAD explantation for patients with evidence of myocardial recovery, despite reverse remodeling occurring in most patients on prolonged LVAD support.¹⁹⁸ Although when patients are carefully selected with optimized LVAD mechanical unloading and standardized HF therapy such as seen in the recent RESTAGE-HF trial, a significantly higher rate of device explantation can be attained.¹⁹⁹ Mann et al.²⁰⁰ previously introduced the idea of “myocardial remission” rather than myocardial recovery to describe patients with a stabilized HF course as well as normalization of the LV at the molecular, cellular, myocardial, and structural levels secondary to reverse remodeling. These patients remain at risk of HF events, even after they regain possession of normal cardiac function such as seen with HF with recovered ejection fraction (HFrecEF). HFrecEF has been defined as a heterogeneous group of patients which is characterized by a less pathological “steady state” associated with improved outcomes compared with patients with reduced or preserved EF.²⁰¹ High rates of HF recurrence have been observed when medical therapy is stopped in those patients. Halliday et al. conducted the TRED-HF trial to assess if withdrawing medications in 51 patients with dilated CMP whose symptoms and cardiac symptoms recover were at increased risk of recurrence.²⁰² Withdrawal of HF medications was associated with 45.7% recurrence of dilated CMP compared to 0% recurrence at 6 months with continuation of therapy. The predictors of relapse included older age, more HF medications, higher global longitudinal strain, higher natriuretic peptides. Despite many of those patients having regained a normal cardiac function and architecture, they may have in fact developed a “HF phenotype deformation memory,” rather than a complete reversal of the HF phenotype, thus vulnerable to face abnormal neurohumoral or hemodynamic loading

conditions, and susceptible to medication cessation. Moreover, the possibility of an “elasticity threshold” has been proposed by Mann et al., potentially explaining why reverse remodeling does not always lead to myocardial recovery.²⁰⁰ Perhaps, the heart that remains within the elastic deformity range is able to demonstrate reversal of the HF phenotype to a normal state, whereas the heart that has sustained irreversible damage after crossing the “elasticity threshold” and transitioning into the “plasticity range” of deformation will not experience myocardial recovery despite demonstrating evidence of reverse remodeling.

The best predictors to date for myocardial recovery in LVAD-supported patients are the time from HF onset and the underlying HF etiology. Additionally, there is evidence that some patients may possess intrinsic biological characteristics that make them better responders (even super responders) or non responders to medical therapies, device therapies such as CRT, and surgical therapeutic options including LVAD support. There is a need to identify histological and molecular findings characterizing the various degrees of myocardial recovery or remission in responders and in patients remaining on the “elasticity curve” compared with patients without evidence of functional recovery. The process of reverse remodeling may lead to two distinct clinical outcomes, namely myocardial remission and myocardial recovery, and the underlying pathophysiological mechanisms explaining their fundamental differences have yet to be fully explained.

Conclusion

Contemporary HF medical and device therapy has led to a growing interest to better understand the phenomena of ventricular reverse remodeling, to improve patient selection for advanced procedures, and provide new insights on mechanistic pathways for research and treatment development. Reverse remodeling may lead to improve myocardial function and, in some patients, to myocardial remission and myocardial recovery, but major abnormalities persist at the molecular level contributing to an elevated risk of cardiovascular events. The earlier the remodeling changes are altered with the use of guideline-directed medical therapy, devices, and with available surgical options, the greater is the chance of reversing adverse LV morphology and architectural changes. The promotion of reverse remodeling may lead to two distinct clinical outcomes namely myocardial remission and less often, complete recovery, but the understanding of their respective pathophysiological mechanisms has yet to be fully understood. There is a need for further research identifying potential mechanistic pathways and optimal patient selection criteria for available therapies promoting reverse remodeling and myocardial recovery.

Funding

None.



Disclosure statement

M.R.M. – payment made to institution from Abbott for consulting. Consulting fees from Mesoblast, Janssen, Portola, Bayer, Triple Gene, and Baim Institute for Clinical Research. Advisory board member for NuPulseCV, Leviticus and FineHeart. J.B. has no conflicts to report.

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